INCH-POUND

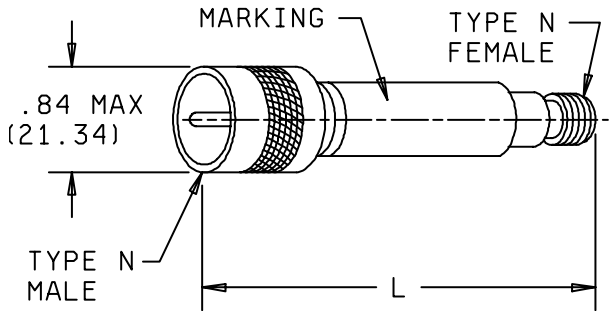
MIL-DTL-3933/18F
7 November 2002
SUPERSEDING
MIL-A-3933/18E
24 April 1985

DETAIL SPECIFICATION SHEET

ATTENUATORS, FIXED, COAXIAL LINE (SERIES N), FREQUENCY RANGE: DC TO 12.4 GHz AND DC TO 18 GHz, LOW POWER

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification and the latest issue of MIL-DTL-3933.



NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Metric equivalents are in parentheses.

FIGURE 1. Dimensions and configuration.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T22:31Z
Check the source to verify that this is the current version before use.

ENGINEERING DATA:

Attenuators for use in general field equipment.

Operating frequency range: DC to 12.4 GHz and DC to 18 GHz.

Test frequency ranges for SWR and attenuation measurements, measured or recorded, when using an Automatic Network Analyzer (ANA) or equivalent test equipment:

100 MHz (or lower) to 12.4 GHz for dash numbers 01 through 20;
100 MHz (or lower) to 18.0 GHz for dash numbers 21 through 24.

Basic test frequencies for sensitive measurements like connector repeatability, temperature sensitivity, power sensitivity and any other measurements where a 'delta' spec is imposed: 6.0 GHz for dash numbers 01 through 20; 8.0 GHz for dash numbers 21 through 24.

Operating temperature range: -55°C to +125°C.

Maximum weight: See table I.

REQUIREMENTS:

Dimensions and configuration: See figure 1 and table I.

Nominal impedance: 50 ohms.

Attenuation: See table I. Stability and sensitivity:

Maximum change in attenuation after:	up to 10 dB, inclusive (dB)	over 10 dB (dB/dB)
temperature change or thermal shock	0.05	0.005
vibration or shock	0.1	0.01
moisture resistance or salt spray	0.2	0.02
peak power	0.05	0.005

Frequency sensitivity, maximum: 0.1dB/dB/GHz.
Temperature sensitivity of attenuation, maximum: 0.0004 dB/dB/°C.

Power: See table I.
Power sensitivity for full input power, maximum: 0.005 dB/dB/Watt.

VSWR: See table I.

Terminal connectors: The series N connectors shall meet the interface dimensions of MIL-PRF-39012. There shall be a male plug on one end and a female plug on the other end.
Connector repeatability, maximum variation in attenuation: 0.02 dB.

Part or Identifying Number (PIN): M3933/18- (and dash number from table I).

Supersession data: See table III.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T22:31Z
Check the source to verify that this is the current version before use.

TABLE I. Electrical characteristics.

Dash number	Attenuation dB			Maximum power input		VSWR 1/ Maximum				Weight (oz) Max	Dimension L (inches) Maximum
	Nominal	Deviation		AV (W) at 25°C (continuously) 2/	Peak 3/ (kW)	DC to 4.0 GHz	4.0 to 8.0 GHz	8.0 to 12.4 GHz	12.4 to 18.0 GHz		
		DC to 12.4 GHz	DC to 18 GHz								
01 N S	3	±0.2	---	2	1.0	1.15:1	1.20:1	1.25:1	---	5	2.70 (68.58)
02 N S	6	±0.2	---	2	1.0	1.15:1	1.20:1	1.25:1	---	5	2.70 (68.58)
03 N S	10	±0.4	---	2	1.0	1.15:1	1.20:1	1.25:1	---	5	2.70 (68.58)
04 N S	20	±0.4	---	2	1.0	1.15:1	1.20:1	1.25:1	---	5	2.70 (68.58)
05 N S	30	±0.8	---	2	1.0	1.15:1	1.20:1	1.25:1	---	5	2.70 (68.58)
06 N S	40	±0.8	---	2	1.0	1.15:1	1.20:1	1.25:1	---	5	2.70 (68.58)
07 N S	50	±1.25	---	2	1.0	1.15:1	1.20:1	1.25:1	---	5	2.70 (68.58)
08 N S	60	±1.25	---	2	1.0	1.15:1	1.20:1	1.25:1	---	5	2.70 (68.58)
09 N S	1	±0.4	---	5	1.0	1.15:1	1.20:1	1.25:1	---	3	2.70 (68.58)
10 N S	2	±0.4	---	5	1.0	1.15:1	1.20:1	1.25:1	---	3	2.70 (68.58)
11 N S	3	±0.3	---	5	1.0	1.15:1	1.20:1	1.25:1	---	3	2.70 (68.58)
12 N S	5	±0.3	---	5	1.0	1.15:1	1.20:1	1.25:1	---	3	2.70 (68.58)
13 N S	6	±0.3	---	5	1.0	1.15:1	1.20:1	1.25:1	---	3	2.70 (68.58)
14 N S	10	±0.4	---	5	1.0	1.15:1	1.20:1	1.25:1	---	3	2.70 (68.58)
15 N S	13	±0.5	---	5	1.0	1.15:1	1.20:1	1.25:1	---	3	2.70 (68.58)
16 N S	20	±0.5	---	5	1.0	1.15:1	1.20:1	1.25:1	---	3	2.70 (68.58)

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T22:31Z
Check the source to verify that this is the current version before use.

TABLE I. Electrical characteristics - Continued.

Dash number		Attenuation dB				Maximum power input		VSWR 1/ Maximum				Weight (oz) Max	Dimension L (inches) Maximum
		Nominal	Deviation			AV (W) at 25°C (continuously) 2/	Peak 3/ (kW)	DC to 4.0 GHz	4.0 to 8.0 GHz	8.0 to 12.4 GHz	12.4 to 18.0 GHz		
				DC to 12.4 GHz	DC to 18 GHz								
17	N S	30		±0.8	---	5	1.0	1.15:1	1.20:1	1.25:1	---	3	2.70 (68.58)
18	N S	40		±0.8	---	5	1.0	1.15:1	1.20:1	1.25:1	---	5	2.70 (68.58)
19	N S	50		±1.0	---	5	1.0	1.15:1	1.20:1	1.25:1	---	5	2.70 (68.58)
20	N S	60		±1.0	---	5	1.0	1.15:1	1.20:1	1.25:1	---	5	2.70 (68.58)
21	N S	3		---	±0.3	2	0.2	1.12:1	1.15:1	1.18:1	1.20:1	3	2.70 (68.58)
22	N S	6		---	±0.3	2	0.2	1.12:1	1.15:1	1.18:1	1.20:1	3	2.70 (68.58)
23	N S	10		---	±0.5	2	0.2	1.12:1	1.15:1	1.18:1	1.20:1	3	2.70 (68.58)
24	N S	20		---	±0.5	2	0.2	1.12:1	1.15:1	1.18:1	1.20:1	3	2.70 (68.58)

1/ VSWR value is for both ends.

2/ Power input is derated linearly to 0 watts at +125°C.

3/ Peak power for a duty cycle of 5×10^{-4}; maximum pulse duration of 5 microseconds.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T22:31Z
Check the source to verify that this is the current version before use.

TABLE II. Supersession data.

PIN M3933/18	Supersedes M3933/	
01	12-1	13-1
02	12-2	13-2
03	12-3	13-3
04	12-4	13-4
05	12-5	13-5
06	12-6	13-6
07	12-7	13-7
08	12-8	13-8

Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Review activities:
Army - MI
Navy - AS, MC, SH
Air Force - 99

Preparing activity:
DLA - CC

(Project 5985-1229-04)

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T22:31Z
Check the source to verify that this is the current version before use.